Forensic Microscopy

Lab 9: Synthetic Fibers

Aritra Bhattacharya
Graduate teaching assistant
ecoevoaritra23@tamu.edu

Jackie Maddox
Undergraduate teaching assistant
jac.r.maddox@tamu.edu

Where can you find the slides?

- https://aritra-evolves.github.io/FIVS215.html
- Canvas

Grab a microscope, set up kohler

Agenda

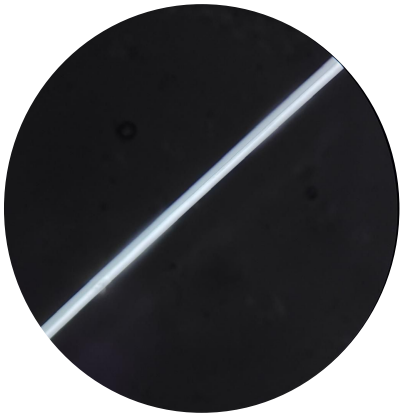
- Identifying Synthetic Fibers
- Michel Levy Chart
- Practice

Sign of elongation
(when kept under cross polars and the compensator plate is fully pushed)

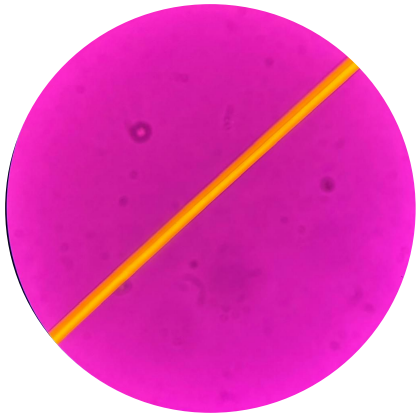
- Low birefringent fibers if appear as blue when kept parallel to the slow axis of compensator (i.e. 1 & 3 quadrant), it is additive (or positive)
- Low birefringent fibers if appear as yellow/amber when kept parallel to the slow axis of compensator (i.e. 1 & 3 quadrant), it is subtractive (or negative)
- If High birefringent fibers show higher order colors when kept parallel to the slow axis of compensator (i.e. 1 & 3 quadrant), it is additive (or positive).
- If High birefringent fibers show lower order colors when kept parallel to the slow axis of compensator (i.e. 1 & 3 quadrant), it is subtractive (or negative).

II	I
III	IV

Sign of elongation
(when kept under cross polars and the compensator plate is fully pushed)



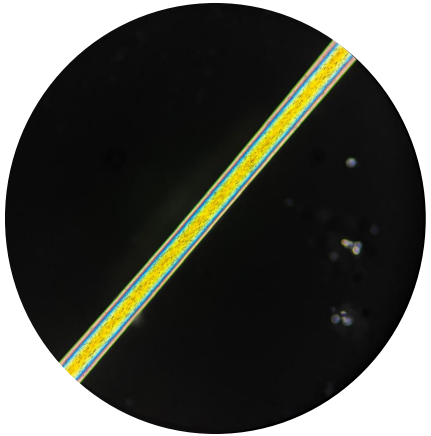
Cross polars
No compensator



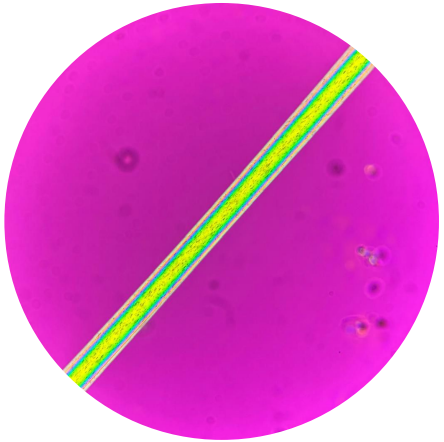
Cross polars
Compensator added

Acetate fiber

Sign of elongation: Subtractive



Cross polars
No compensator



Cross polars
Compensator added

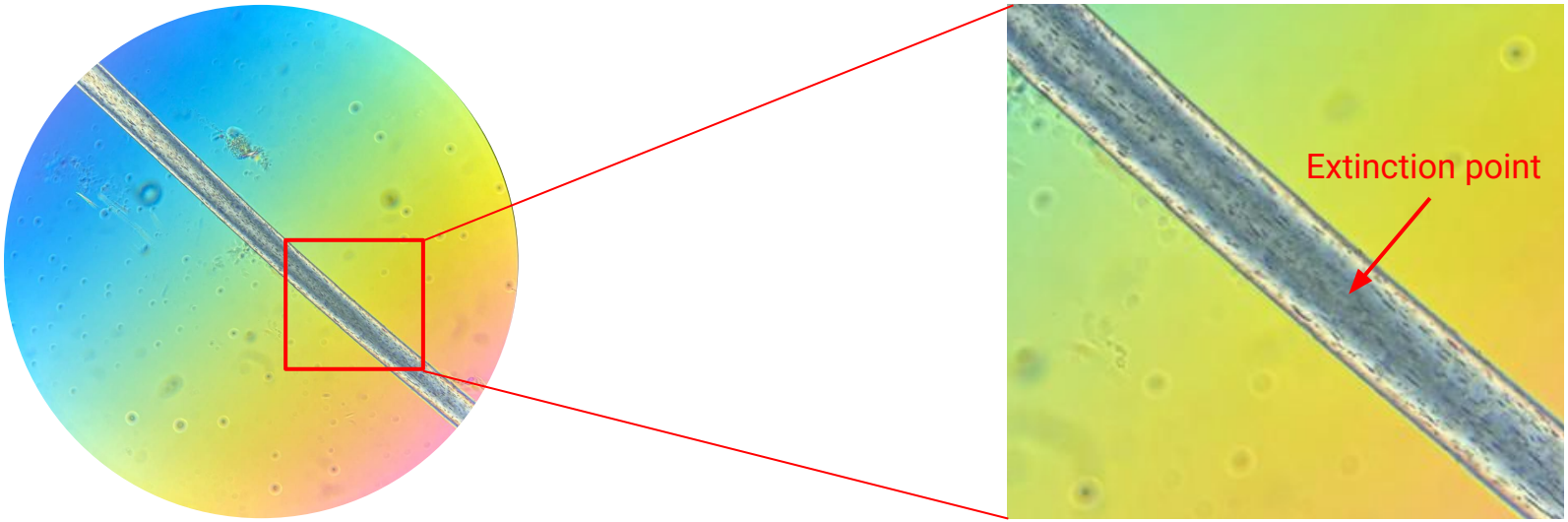
Olefin fiber

Sign of elongation: Additive

Becke Lines

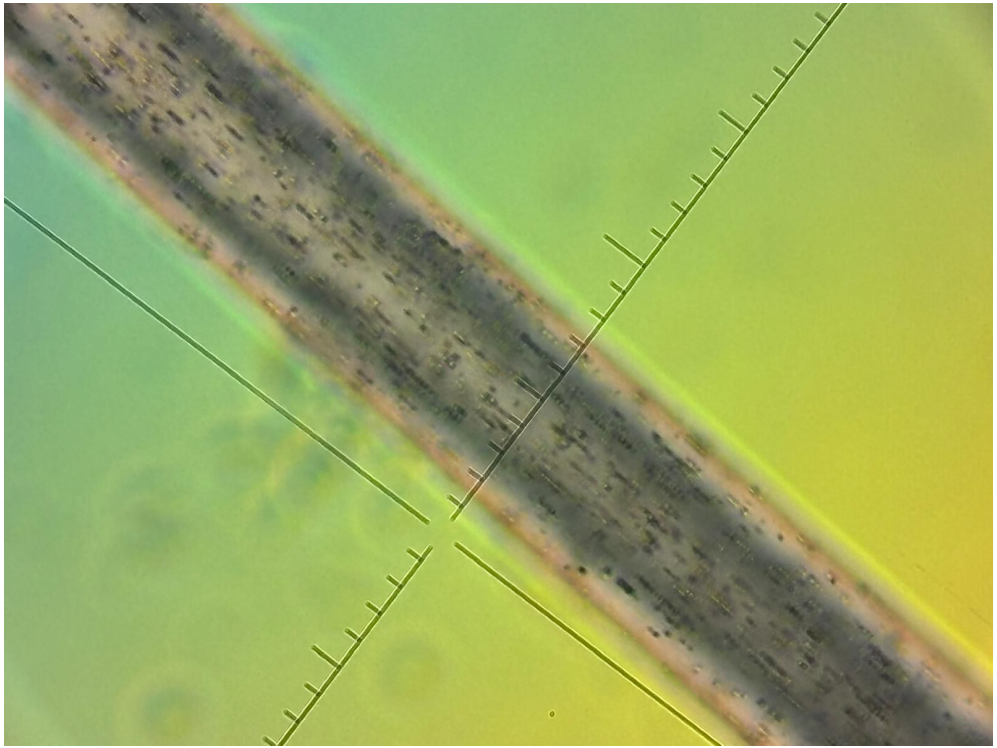
- Halo moves toward higher refractive index.
- If halo moves inward to the fiber, the fiber has higher RI than the mounting media and vice versa

Birefringence *(with quartz wedge)*



- Keep the fiber perpendicular to the slow axis of the quartz edge (i.e.: diagonally in 2 and 4 quadrant).
- Insert the quartz edge slowly to find an extinction point (where the thicker region of the fiber becomes grey/black).
- Observe the background color around that extinction point

Birefringence *(with quartz wedge)*



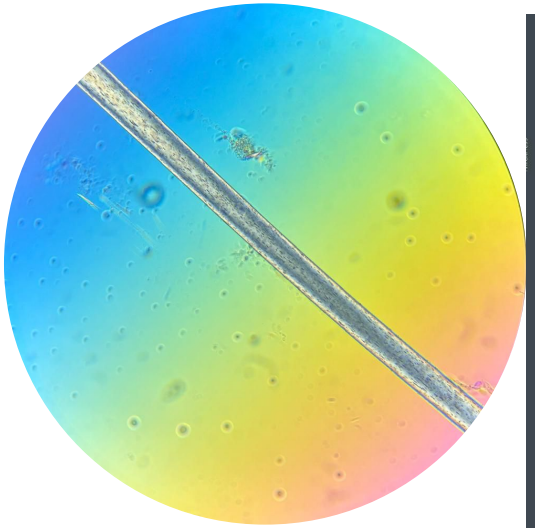
- Measure the thickness of your fiber, (Use your micrometry lab calculations)

4X: ~50um
10X: ~20um
40X: ~5um
63X: ~3um

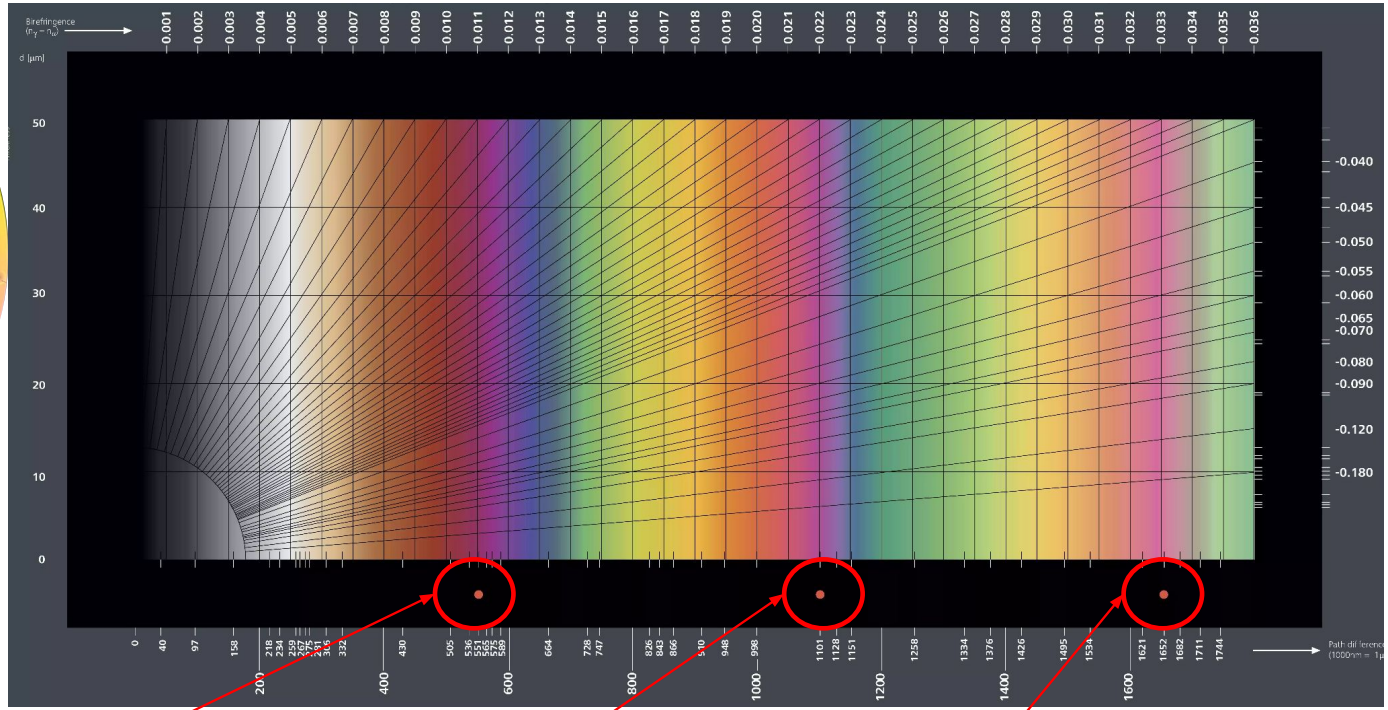
Olefin fiber under 40X
(thickness roughly 35um)

Birefringence *(with quartz wedge)*

Match the thickness of fiber and the color around extinction point on Michel Levy chart



Olefin fiber under 40X
(thickness roughly 35um)



Practice

1. Measure the Thickness of your Fiber
2. Insert Full Wave Plate
    a. Observe if it's Additive or Subtractive
    b. Blue or Amber
3. Find the Refractive Index Range
4. Find the Relief
    a. In Parallel and Perpendicular
5. Put Fiber in Subtractive Orientation (amber)
6. Cross Polars
7. Insert Quartz Wedge and Count the Orders
8. Use the Michel Levy Chart to Find Birefringence

Fibers to Look At

Pick at Least 4

1. Acetate
2. Polyester
3. Rayon
4. Acrylic/Orlon
5. Nylon
6. Olefin

See you next week!